SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I J. Lakshmi Devi (192584300) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Lakshmi Devi

Signature of the student:

90/100

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

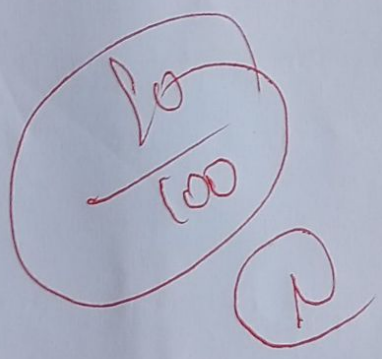
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
   a. Show differences and similarities
      i.Map how each affects the Data structure and Security level
      ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
   a. Show relationships between these concepts
   b. Analyze how each contributes to secure communication
   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
   a. Show how each concept is mathematically connected
   b. Map their role in encryption/decryption
   c. Analyze which concept is most critical for public key cryptography and justify

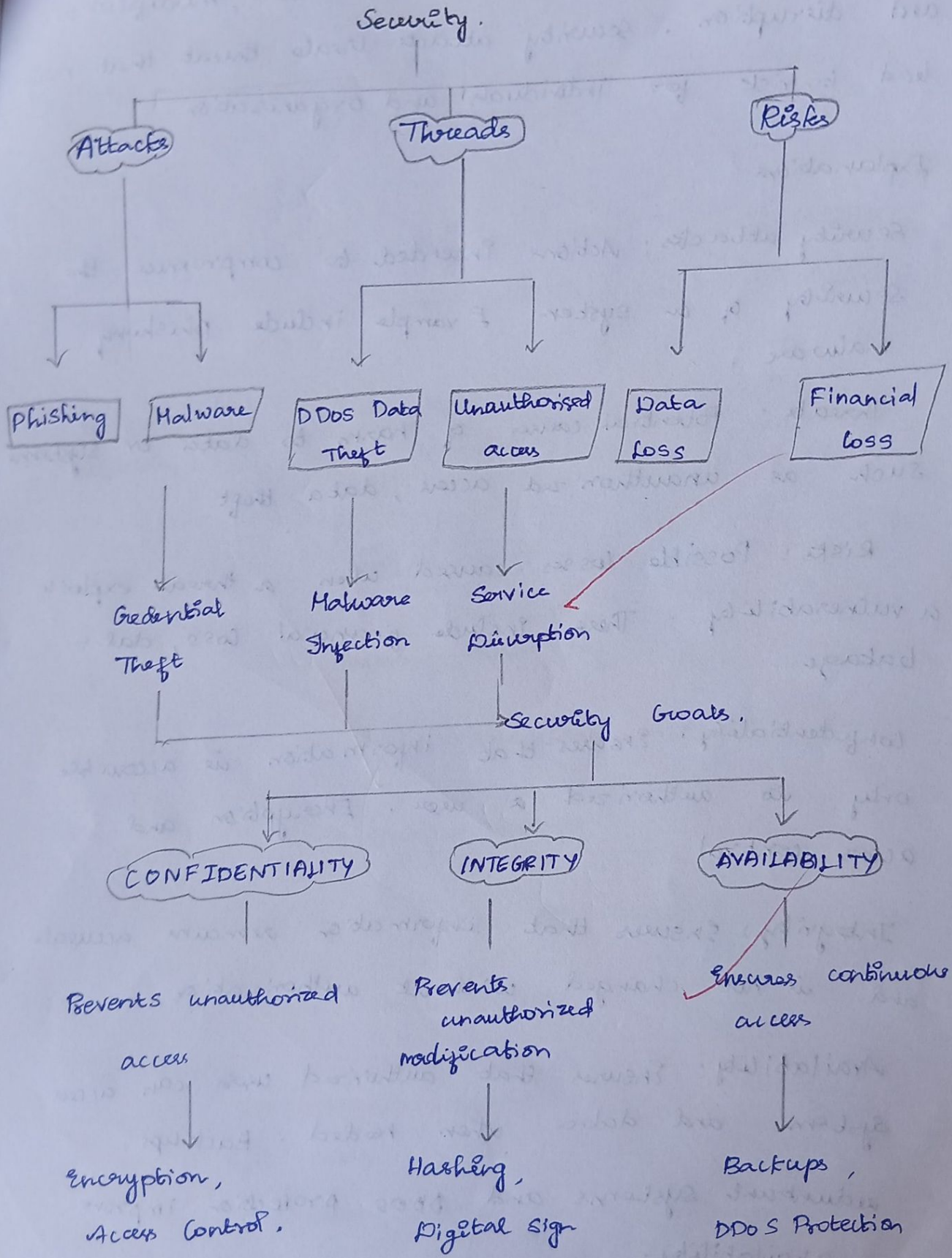
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



LO1 Assessment tool - 02.

1. Security Attacks, Threats, Risks and Security Goals.



Introduction:

Information security protects data and computer systems from unauthorized access, modification and disruption. Security attacks create threat that may lead to risk for individuals and organization

Explanations

Security attacks: Actions intended to compromise the security of a system Example include phishing, malware.

Threats: Potential causes of harm to data or systems such as unauthorized access, data theft

Risks: Possible losses caused when a threat exploits a vulnerability. These include financial loss, data leakage.

Confidentiality: Ensures that information is accessible only to authorized users. Encryption and access control

Integrity: Ensures that information remains accurate and is not changed without authorization,

Availability: Ensures that authorized users can access systems and data when needed. Backups, redundant systems and DDOS protection improve availability.

SUBSTITUTION TECHNIQUE

Caesar Cipher

Monoalphabetic Cipher

Vigenere Cipher

One fixed shift key

One - to - one substitution

Multiple substitution

Simple complexity

Moderate complexity

Higher Complexity

HILL CIPHER

Matrix - based key

Higher complexity

Stronger security

MOST SECURE OF THESE

HILL CIPHER

Uses matrix operations and multiple letters together, making simple frequency analysis harder.

Introduction:

Substitution ciphers are classical encryption method in which plaintext letters or groups of letters are replaced with other letters or symbols.

techniques:

Caesar Cipher: Each letter is shifted by a fixed number of positions. It is simple to implement but easy to break.

Monoalphabetic Cipher: Each plain text letter is replaced with a fixed substitute letter. It has more possible keys than Caesar Cipher, but letter-frequency analysis.

Vigenere Cipher: Uses a repeating keyword to apply different shifts to plain text letters.

Hill Cipher: Uses matrix multiplication and a matrix key to encrypt groups of letters. It provides greater complexity and can hide individual-letter frequency pattern

Real-world relevance

Classical substition ciphers are useful for learning the foundation of encryption, key management and cryptanalysis. Modern system use stronger algorithms such as AES to protect sensitive data.